TextMamba3D

Overfitting Analysis & Improvement Plan

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TextMamba3D is a text-guided 3D brain tumor segmentation framework based on a unified Mamba (State Space Model) architecture. This report systematically analyzes the overfitting risks identified during training and proposes a three-phase improvement plan covering experimental design, regularization, and model slimming.

Metric	TextMamba3D	SegMamba	TextBraTS Paper
Total Params	~180M	~23M	N/A
Trainable Params	~70M	~23M	N/A
Training Samples	295	~877	220
Param/Sample Ratio	237K:1	26K:1	-
Dataset	TextBraTS 369	BraTS2023 1251	TextBraTS 369
Current Best Dice	0.637	-	0.853

Contents

1. Core Problem Diagnosis

1.1 Overfitting Risk (Critical)

TextMamba3D uses 3x the trainable parameters to fit 1/3 of the data compared to SegMamba. The parameter-to-sample ratio is ~9x that of SegMamba, creating a structurally over-parameterized scenario.

Dimension	TextMamba3D	SegMamba	Ratio
Trainable Params	~70M	~23M	3x
Training Samples	295	~877	1/3x
Param/Sample	237K:1	26K:1	9x

Architecture Component Breakdown:

Component	Module	Params	Key Design
Modality Grouping	PatchEmbed3D	~2M	T1+T1ce / T2+FLAIR
Image Backbone	CrossScanBiMamba3D x4	~44M	6-direction scan
Text Encoder	PubMedBERT + Mamba	~110M (109.5M frozen)	Last 2 layers unfrozen
Fusion	FiLM + CrossAttn + MambaFusion	~7M	Dual-path fusion
Uncertainty Gating	UncertaintyGating	~1M	Noise suppression
Decoder	MambaDecoder + DS Heads	~16M	Deep supervision

1.2 Experimental Design Flaws

Fatal Issue: val == test. In `_split_cases()`, both `split='val'` and `split='test'` return the same 74 cases. Best checkpoint selection is based on these same cases, making all reported metrics optimistically biased.

The TextBraTS original paper uses a proper 220/55/94 (train/val/test) three-way split with 94 independent test cases. TextMamba3D has no independent test set and no cross-validation.

```
if split == 'train': indices[:train_size]
else: # val or test -- returns same data!
indices[train_size:]
```

1.3 Insufficient Regularization

Technique	Current Value	Problem
Dropout	0.0 (entire model)	70M params with zero stochastic regularization
Weight Decay	1e-5	Too small to be effective
PubMedBERT	Last 2 layers unfrozen (+14M)	14M text params on 295 samples

Technique	Current Value	Problem
No-text Ratio	0.15	Could be higher for better regularization

Only existing regularization: data augmentation (7 types), deep supervision, gradient clipping (max_norm=1.0), and early stopping (patience=30).

1.4 Hardware Limitations (RTX 4060 8GB)

Constraint	Current	Ideal	Impact
Batch size	1	4~8	Noisy gradients, unstable training
Patch size	64^3	96^3~128^3	Only ~3% of original volume (240x240x155)
Contrastive loss	Disabled	batch>1 needed	Designed but unused
Training speed	Very slow	-	Hyperparameter search infeasible

patch_size=64^3 is the most critical hardware constraint. Brain tumor segmentation requires global context (tumor location, ventricle relationship, edema extent). This partially explains the Dice plateau at 0.63 -- not just overfitting, but also underfitting due to insufficient spatial context.

2. Performance Benchmarks

Same Dataset (TextBraTS 369 Cases)

Method	ET Dice	WT Dice	TC Dice	Avg Dice
3D-UNet	80.4%	87.3%	81.6%	83.1%
nnU-Net	82.2%	87.5%	82.6%	84.1%
SegResNet	80.9%	88.4%	82.3%	83.8%
SwinUNETR	81.0%	89.5%	80.8%	83.8%
TextBraTS (paper)	83.3%	89.9%	82.8%	85.3%
TextMamba3D (ep.11)	-	-	-	62.3%
TextMamba3D (best ep.62)	-	-	-	63.7%

TextMamba3D uses 295 training samples (vs 220 in the original paper) but achieves significantly lower performance, confirming that overfitting + hardware constraints are actively harming generalization.

Expected Dice After Improvements

Scenario	Expected Avg Dice	Basis
Current (no changes)	0.65~0.70	Historical best 0.637, val=test bias
Phase 2 (regularization)	0.75~0.80	Dropout + weight decay + freeze BERT
Phase 2+3 (+ slimming + better GPU)	0.80~0.85	Approaching nnU-Net level
Theoretical ceiling	~0.85	TextBraTS paper SOTA

3. Three-Phase Improvement Plan

Principle: Fix evaluation first, then regularization, then architecture. Never tune the model while the evaluation system is flawed.

3.1 Phase 1: Fix Experimental Design

Goal: Make evaluation trustworthy. No model code changes.

Task	File	Details
Separate val/test	brats_textbrats_dataset.py	Ref: original paper 220/55/94 split
Or: implement 5-fold CV	train.py + dataset	369 cases fully utilized, report mean +/- std
Add train/val loss curves	train.py	TensorBoard logging for overfitting detection
Run baseline	-	Full training with current model as control

3.2 Phase 2: Regularization Enhancement

Goal: Reduce trainable params, add regularization. Expect Dice 0.75~0.80.

Change	Current	Target	Effect
Dropout	0.0	0.1~0.2	Stochastic regularization in Mamba blocks
Weight Decay	1e-5	1e-2~5e-3	Stronger L2 penalty
Freeze PubMedBERT	Last 2 layers unfrozen	Fully frozen	-14M trainable params
Or: LoRA (r=8)	14M unfrozen	+0.3M LoRA	97% fewer text params
No-text ratio	0.15	0.25~0.30	Stronger modality regularization

Trainable params: ~70M -> ~56M (freeze BERT) or ~56.3M (LoRA).

3.3 Phase 3: Model Slimming (Only If Phase 2 Still Overfits)

Goal: Match model capacity to data scale. Expect Dice 0.80~0.85.

Change	Effect	Risk
Base dim: 96 -> 48	Image encoder ~44M -> ~11M	May reduce expressiveness
Depths: [2,2,2,2] -> [1,1,2,1]	~50% param reduction	Low risk, keep bottleneck depth=2
Remove Cross-Attention	Save ~3-4M	Needs ablation validation
Scan directions: 6 -> 3	~50% less compute	Aligns with SegMamba's design

Trainable params: ~56M -> ~20M.

4. GPU Requirements & Selection

VRAM Estimation (with AMP + Gradient Checkpointing)

Phase	Trainable	Patch	Batch	Est. VRAM	Min GPU
Phase 2 (dim=96)	~56M	64^3	1	~7 GB	8 GB (4060)
Phase 2 (dim=96)	~56M	96^3	1	~14 GB	16 GB (T4)
Phase 2 (dim=96)	~56M	96^3	2	~24 GB	24 GB (3090)
Phase 2+3 (dim=48)	~20M	96^3	2	~12 GB	16 GB (T4)
Phase 2+3 (dim=48)	~20M	96^3	4	~21 GB	24 GB (3090)
Phase 2 (dim=96)	~56M	128^3	2	~38 GB	40 GB (A100)

Safety margin: use only 80-85% of nominal VRAM. If estimated need is 20GB, choose a 24GB GPU. Full VRAM utilization causes OOM crashes, cuDNN tuning failures, and fragmentation.

Recommended Configurations

Plan	GPU	Config	Cost
Best value	T4 16GB	Phase 2+3, patch 96^3, batch 2	Colab Free
Balanced	RTX 3090 24GB	Phase 2, patch 96^3, batch 2	AutoDL ~3 CNY/hr
Best experience	A100 40GB	Phase 2, patch 128^3, batch 2	Colab Pro ~$12/mo

Local RTX 4060 8GB: code development, debugging, smoke tests only (--max-samples 50). Formal training and ablation experiments on remote 16GB+ GPU.

5. Text Guidance Value Assessment

Current Evidence

Metric	Value
With text Dice	0.6233
Without text Dice	0.5959
Absolute gain	+2.74%
Measurement point	Epoch 11/150 (far from convergence)
Statistical test	None (single split, single run)

Why Current Evidence Is Unreliable

- Model Dice is 0.62 vs baseline 0.84 -- measuring gains on severely underfitting model is meaningless

- Text branch adds ~21M trainable params -- the +2.7% could be parameter capacity, not semantic information

- TextBraTS original paper showed only +1.2% gain (85.3% vs 84.1%) with simpler fusion

- No statistical significance test (single split, single run)

Required Experiments to Validate

- Fair ablation: remove text branch, add equivalent params to image encoder, compare at same total params

- Converge first: measure text gain only when Dice > 0.80

- Multiple runs: at least 3 seeds, report mean +/- std

- Statistical test: paired t-test or Wilcoxon signed-rank test

6. Idea2Paper RAG System Overview

The Idea2Paper RAG system retrieves relevant research patterns from a knowledge graph of 8,285 ICLR 2025 papers using a three-path fusion retrieval strategy.

Knowledge Base

Entity	Count	Source
Papers	8,285	ICLR 2025
Ideas	8,284	Extracted from papers
Patterns	124	Clustered research paradigms
Domains	98	Research domains
Edges	485,508	Directed relationships

Three-Path Fusion Retrieval

Path	Weight	Flow
Path 1: Idea->Idea->Pattern	0.4	User idea matches 8,284 ideas -> top-20 -> trace Patterns
Path 2: Idea->Domain->Pattern	0.2	Best idea -> graph traversal -> find applicable Patterns
Path 3: Idea->Paper->Pattern	0.4	User idea matches 8,285 titles -> top-20 -> trace Patterns

Each path uses two-stage coarse-to-fine ranking: Jaccard token overlap (top-100) then embedding cosine similarity (re-rank). Embedding: Gemini text-embedding-004. Vector store: raw NumPy .npy (brute-force exact search, sufficient for 8K papers).

Novelty Check

Parameter	Value
High risk threshold	cosine >= 0.88
Medium risk threshold	cosine >= 0.82
Action on high risk	Auto-pivot (max 2 times)
Index text	title + domain + idea + pattern_details

7. Key Concepts Reference

7.1 LoRA (Low-Rank Adaptation)

Parameter-efficient fine-tuning: W' = W + A x B, where A (d x r) and B (r x d), r << d. Freezes original weights, trains only low-rank matrices. For a 7B model: full fine-tuning = 7B params, LoRA (r=16) = ~20M params (0.3%).

Relevance to TextMamba3D:

Approach	Trainable Params	Overfitting Risk
Current: unfreeze BERT last 2 layers	+14M	High
Fully freeze BERT	+0	Lowest
LoRA (r=8) on BERT	+0.3M	Low (recommended)

7.2 Domain Shift

Distribution mismatch between training and deployment data. In medical imaging: scanner differences (GE vs Siemens), acquisition protocols (1.5T vs 3T), patient demographics, and annotation standards.

Type	Definition	Example
Covariate Shift	P(X) changes, P(Y|X) unchanged	Different MRI scanners
Label Shift	P(Y) changes	Different tumor type ratios by region
Concept Shift	P(Y|X) changes	Different radiologist annotation standards

Current priority: overfitting is more urgent than domain shift. Address domain shift after model converges (Dice > 0.80). But adding Domain Randomization (brightness/contrast/gamma) during augmentation is low-cost and can be done now.

7.3 FiLM Modulation

Feature-wise Linear Modulation: output = gamma * features + beta, where gamma (scale) and beta (shift) are predicted by conditional input (text). Channel-level global modulation, O(C) complexity -- much lighter than Cross-Attention O(n*m).

Dimension	FiLM	Cross-Attention
Granularity	Channel-level (global)	Token/pixel-level (local)
Compute	O(C), very light	O(n*m), heavier
Expressiveness	Global scale/shift	Fine-grained spatial alignment
Use case	'Overall style adjustment'	'Which pixel maps to which word'

8. Action Checklist

Local (RTX 4060 8GB) -- Code Changes

#	Task	Details	File
1	Modify _split_cases()	Three-way split or 5-fold CV	brats_textbrats_dataset.py
2	Add dropout=0.1	All Mamba blocks	mamba_block.py, encoder_3d.py
3	Update YAML config	weight_decay=1e-2, unfreeze=0, no_text=0.25	textbrats.yaml
4	Smoke test	--max-samples 50	train.py

Remote GPU (16GB+) -- Formal Training

#	Task	Details	Environment
5	Phase 2 training	patch 96^3, batch 2+, full 150 epochs	Remote GPU
6	Evaluate Phase 2	Check if overfitting persists	train/val curves
7	Phase 3 (if needed)	dim=48, reduced depths	Remote GPU
8	Text guidance ablation	After Dice > 0.80, fair comparison	Remote GPU

Target: Dice 0.80+ (matching nnU-Net 0.841), then validate text guidance gain.

References

[1] TextBraTS: Text-Guided Volumetric Brain Tumor Segmentation (MICCAI 2025) -- arxiv.org/abs/2506.16784

[2] SegMamba: Long-range Sequential Modeling Mamba for 3D Medical Image Segmentation -- arxiv.org/abs/2401.13560

[3] LoRA: Low-Rank Adaptation of Large Language Models -- arxiv.org/abs/2106.09685

[4] FiLM: Visual Reasoning with a General Conditioning Layer -- arxiv.org/abs/1709.07871

[5] Mamba: Linear-Time Sequence Modeling with Selective State Spaces -- arxiv.org/abs/2312.00752